Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Gewirtz, Alan M. M.D.		
Project Title:	Effect of Deep Space Radiation on Human Hematopoietic Stem Cells		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104	**Congressional District:**	2
Comments:	PI passed away in 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sutherland, Betsy (Brookhaven National Laboratory)		
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	Astronauts on long term missions in deep space will be placed at risk from a variety of hazards. Some of these are known while others may be anticipated. Damage to hematopoietic stem cells as a result of radiation exposure is as an example of the latter. Our long term goal is to identify and quantitate the risks of deep space radiation to the human hematopoietic system, with particular emphasis on the hematopoietic stem cell. Stem cells are the ultimate source of both the blood and immune systems and damage to these cells could have grave immediate, and long term consequences. At the same time, because these cells can be readily removed from the body, manipulated, and stored, they are also unique candidates for countermeasures that might obviate, or totally negate, damage incurred to them. Accordingly, this project will have three specific aims which support our long term goal and these are to: 1] Investigate the cellular consequences of exposing human hematopoietic stem (HSC)and progenitor (HPC)cells to an environment which simulates the radiation environment of deep space; 2] Examine the molecular consequences of exposing human		

Task Description:	hematopoietic stem cells to an environment which simulates the radiation environment of deep space. This aim has two purposes. If radiation leads to degradation of hematopoietic cell function it will clearly be of interest to look for the molecular lesions potentially responsible for such damage. Alternatively, more long term, but initially occult damage may also be induced. The consequences of such damage could lead either to a complete failure of hematopoiesis (aplastic anemia) or the development of hematologic malignancies. Identification of such damage is therefore important; and 3] Design potential countermeasures to obviate or negate cellular and molecular damage discerned during the course of carrying out Aims 1 and 2. We propose both simple and more complex solutions to problems that might be identified during the course of this study. We suggest that prophylactic (pre-flight) harvest and storage of astronaut stem cells might be a safe, effective, and relatively inexpensive mechanism for countering long term damage to cells of the hematopoietic systems. Countermeasures which might prove effective in combating damage encountered during flight will also be developed and explored for their utility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of effective radiation countermeasures could have significant impact on earth for civilians and military personnel alike. With regard to the civilian population, it is quite conceivable that results we obtain will be relevant to patients undergoing cancer chemotherapy and radiation therapy. In this regard it is possible that our studies will provide reagents and strategies for helping to protect normal tissues from the collateral damage of anticancer treatments. It is also possible that results we generate will be relevant to radiation workers, and members of the armed forces who may be exposed to radiation during the course of carrying our their respective duties.
Task Progress:	The deep space radiation environment is composed of both low-, and high-, LET radiation sources. High-LET radiation particles (HZE) are particularly dangerous because, by direction ionization of DNA, or radiolysis of water, they induce free radical formation that causes relatively non-repairable double strand breaks (DSB) and other complex damages to cellular DNA. The risk to flight crews exposed to HZE on extended duration voyages, such as the proposed Mars mission, has not been quantitated. Crew safety, and mission planning, dictate that such studies be done. We are examining the effects of low and high-LET radiation on human hematopoietic stem and progenitor cell function using normal human CD34+ cells obtained from consenting donors. Low-LET radiation was derived from a 137Cs source while HZE, primarily in the form Fe, Ti, Si ions were generated by the Brookhaven National Laboratory (BNL) relativistic heavy ion collider-alternating gradient synchrotron (RHIC-AGS). Cells were prepared at UPENN, and shipped to BNL by overnight courier. The cells were then irradiated, with or without candidate radioprotectants, and aliquots of cells were then analyszed for DNA damage, and colony forming unit (CFU) function. Our initial studies indicated that human CD34+ cells demonstrated a dose dependent sensitivity to exposures as low as 15 cGy of both radiation types. The effects of HZE particles were more severe as shown by the dramatic decrease in assayable BFU-E, CFU-E, CFU-GM and CFU-GEMM colonies. PMH treatment prior to irradiation, enhanced CFU formation by all lineages from 2-4 fold in the dose range of 15-50 cGy (137Cs) as compared to untreated cells. When doses increased to 70 cGy, PMH still displayed significant radioprotective effects in BFU-E and CFU-GEMM assay, as much as 2-3 fold, indicating some lineage specificity to its protective abilities. EUK-134 displayed significant radio-protection between 15-30 cGy as well and was slightly additive when combined with PMH. CFU were increased several fold compared to untreated controls in the dose range of 30-50 cGy of Fe. At 50 cGy, CFU-E, and CFU-GM were no longer protected. A decrease in the levels of phosphorylated histone H2A.X, a sensor of DSB, was observed in cells treated with PMH, and EUK-134, and correlated with the radioprotection observed. Immunochemical staining shows an increase in repair proteins unique for DSB, such as Ku70/80, MRE-11 and Rad-50 which were never before documented in these cells. Finally, the number of DNA ends labeled with dUTP-(FITC) terminal transferease enzyme was also increased. Altogether, these observations suggest that between 15-50 cGy, PMH and EUK-134 prevent DNA damage and DSB formation, thereby preserving CFU compared to untreated controls. Protective effects are lost when doses were greater than 50 cGy. We conclude that PMH and EUK-134 is effective against both low and high-LET, and that similar radioprotective mechanisms may be triggered in cells by each. Experiments to address the precise mechanisms of radioprotection by these compounds are ongoing in our labs, and will hopefully assist in the development of even more efficient radioprotective agents.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Presentation	Gurudutta U Gangenahalli, Ph.D. 1, John M Millholland, Ph.D. 1, "Development of Countermeasures to Protect Human " Jul-2004
Presentation	Gurudutta U Gangenahalli, Ph.D. 1, John M Millholland, Ph.D. 1, "Modification in the induction levels of DNA double strand " Jul-2004